

Viva Prep — Everything to Remember

The prototype, the problem, and the solution — one page per topic, plain language

1. The basics — say these without hesitating

- **Team:** SANKALP.
 - **Prototype:** INAYAT — means “care” in Urdu. Named that on purpose: it exists to give every patient the caring attention a rushed visit can't.
 - **Problem Statement:** SIH26-CC-0032 — Patient Case-Taking Software.
 - **What it is, in one line:** a tool that turns what a patient says (typed, spoken, or an old prescription photo) into a clean, structured history the doctor reads before walking in — with safety checks that run before anything else.
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2. The problem — in plain words

- A doctor in an Indian OPD gets about **two minutes** with each patient.
 - But **70–80%** of a correct diagnosis comes from the patient's history alone — before any test.
 - So the two minutes doctors have least of is exactly the two minutes that matters most.
 - Patients bring torn, unstructured old prescriptions no one has time to read properly.
 - Elderly, rural, and low-literacy patients get left out of most digital health tools.
 - Doctors end up asking the same basic questions over and over, wasting the little time they have.
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3. The solution — five things that make it real

- **Multilingual voice** — English, Hindi, Telugu, both text and speech, so language is never a barrier.
 - **Adaptive questioning** — a real follow-up question engine (SOCRATES framework): a cough gets asked about phlegm, a rash gets asked about spread. One question at a time, always skippable.
 - **Safety-first triage** — a red-flag scanner runs before anything else, catching genuine emergencies (chest pain, unconsciousness, severe bleeding) instantly, with zero AI call needed.
 - **Smart document reading** — OCR reads old prescriptions and lab reports, pulls out medicines, dates, diagnoses, and flags abnormal lab values against the report's own stated range.
 - **AYUSH-aware history** — real Dashavidha Pariksha (the ten-fold Ayurvedic examination) questions for Ayurvedic consultations, not just “Ayurveda” relabeled onto a generic form.
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4. How it's built — the real stack

- **Frontend:** plain HTML/CSS/JavaScript, no framework, no build step — loads fast on a cheap phone with a weak connection.
 - **Backend:** Python, FastAPI, SQLite.
 - **Reasoning:** Claude (Anthropic) or Groq for live language reasoning, backed by our own TF-IDF clinical-guideline safety check that runs independently of either.
 - **OCR:** Tesseract, reading prescriptions/lab reports into plain text, then regex-based extraction (medications, dates, diagnoses, lab values).
 - **Voice:** Bhashini (India's government ASR/TTS service) live, with a fully offline fallback (PocketSphinx + espeak-ng) when Bhashini isn't reachable.
 - **Health ID:** ABDM (Ayushman Bharat Health Account) — real OTP-based linking against India's actual government health-ID API, not a mock.
 - **The agent pipeline (real module names, not a diagram invented for slides):** intake.py (red-flag scan) → socrates_intake.py (follow-up questions) → history_intake.py (structures the narrative) → triage.py (proposes a priority level) → verify.py (guideline-evidence check, can escalate) → referral.py (decides if a doctor must manually review).
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5. Numbers to have ready

- **70–80%** — share of a correct diagnosis that comes from history alone (classical clinical teaching).
- **~2 minutes** — average Indian primary-care consultation length (BMJ Open, 2017, 67-country study).
- **3 languages** — English, Hindi, Telugu (Telugu chosen since the example district, Kurnool, is Telugu-speaking).
- **388 automated tests**, all passing, zero known regressions as of today.
- **Zero** — the API cost of every safety-critical check (red-flag scan, guideline evidence, priority escalation).
- **10** — the Dashavidha Pariksha, the real ten-fold Ayurvedic examination framework the AYUSH questions are checked against.

Q: What's the real differentiator, if you only get one line?

- “Every safety check — the emergency catch, the priority level, the guideline citation — runs with zero API cost and keeps working even if the AI service is down. That's what lets this scale to a real, high-volume government hospital, not just stay a demo.”
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6. Tough questions, short honest answers

Q: Why not just let the patient write it on paper?

- Paper doesn't catch danger — nothing happens until a doctor reads it. This checks the moment it's typed or spoken.
- Not everyone can write it well — elderly, low-literacy, or non-native-language patients. This lets them just speak or show a photo instead.
- Paper is raw notes — the doctor still has to organize it themselves, in the two minutes they don't have. This hands it over already organized the way a doctor thinks.

Q: What dataset did you train it on?

- None — no model was trained. The red-flag scan and follow-up questions are fixed rules, not learned from data.
- The reasoning is a real, already-trained model (Claude/Groq) doing live reasoning — not something built from scratch.
- The only “data” is a small clinical-guideline reference set written and checked by the team — openly not clinically certified yet.

Q: Does it work fully offline?

- The page itself needs one internet load, like any website.
- Once loaded: if the AI can't be reached, it doesn't crash — it safely falls back to a conservative, evidence-gated priority level instead of guessing.
- Voice has a real offline fallback (PocketSphinx/espeak-ng) if the live government ASR service is unreachable.

Q: Can a patient delete their data?

- Not yet, honestly — there's no delete button today. A doctor can correct or amend a case, but records aren't destroyed.
- A real, known gap — not hidden.

Q: Is the code public? Can we check it?

- Yes — real GitHub repo, real commit history: github.com/azlanabyssal-cloud/carepilot

Q: Why does a mild symptom show a serious-sounding priority?

- It only raises the level when it finds real evidence (a matched clinical guideline, shown on screen with its similarity score) — with no match, it stays cautious rather than guessing, and never quietly lowers a level on a weak match.
- That's a deliberate safety design, not a confidence score.

Q: Isn't this just Practo or 1mg?

- No — those book appointments and sell medicine. This is a pre-visit history-taking and safety-triage tool that hands a doctor a ready summary before the patient even walks in — a different problem entirely.

Q: Who pays for this? What's the business model?

- Built for government/public-hospital OPD deployment first — the zero-API-cost fallback exists specifically so it works without a recurring per-query AI bill, the real constraint a public hospital faces.

Q: What about the Digital Personal Data Protection Act, 2023?

- Consent is real and enforced — checked client-side for a clear message, and again server-side (the request is rejected without it), not just a UI checkbox.
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7. Honest gaps — say these plainly if asked, don't dodge

- Every result is labeled “AI-drafted, not yet reviewed by a physician” — by design, a doctor always makes the final call, this never replaces that.
 - No delete-my-data button yet — a real, known gap, not hidden.
 - A separate CV image-triage model exists in the codebase (tested against synthetic images) but isn't wired into the live endpoints yet — named honestly as a build-roadmap item, not claimed as shipped.
 - OCR'd document findings (medications, abnormal labs) are shown to the doctor but don't yet change the `priority_level` automatically — the physician remains the backstop for that judgment either way.
 - The clinical-guideline reference set is small and team-written, openly not a certified medical database.
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8. If you forget everything else — the ten lines

- Team SANKALP. Prototype INAYAT — “care” in Urdu. PS SIH26-CC-0032, Patient Case-Taking Software.
- A doctor gets two minutes; history gives 70–80% of the diagnosis. That's the real problem.
- Three ways in: type, speak (EN/HI/TE), or photograph an old prescription.

- Adaptive follow-up questions, a real safety-first red-flag scanner, OCR with abnormal-lab flagging, AYUSH-aware history — not just a chatbot.
- Every priority level is evidence-gated: a matching guideline can raise it, a weak match never lowers it.
- Zero API cost for every safety-critical check — keeps working even if the AI service is down.
- Real ABDM health-ID linking, real OTP, real government API — not a mock.
- 388 automated tests passing. Code is public on GitHub.
- Every AI-drafted summary says so on screen — helps the doctor decide, never replaces the doctor.
- Known gaps said out loud: no data-deletion yet, CV model not wired in yet, OCR doesn't drive priority yet.